

FYMSc(CA) Sem II Web Technologies Practical

SLIP 1 SOLUTION

♦ Q1: JavaScript – Check Driving Eligibility

```
<!DOCTYPE html>
<html>
<body>
<h3>Driving Eligibility</h3>

Age: <input type="number" id="age">
<button onclick="check()">Check</button>

<p id="result"></p>

<script>
function check() {
    var age = document.getElementById("age").value;
    if(age >= 18)
        document.getElementById("result").innerHTML = "Eligible for driving";
    else
        document.getElementById("result").innerHTML = "Not eligible";
}
</script>

</body>
</html>
```

♦ Q2: PHP – Count Vowels & Occurrences

HTML Form

```
<form method="post" action="vowel.php">
Enter String: <input type="text" name="str">
<input type="submit">
</form>
```

PHP Script (vowel.php)

```
<?php
$str = $_POST['str'];
$str = strtolower($str);
$vowels = ['a','e','i','o','u'];
$count = 0;
foreach($vowels as $v){
    $c = substr_count($str, $v);
    echo "Count of $v = $c <br>";
    $count += $c;
}

echo "Total Vowels = $count";
?>
```

SLIP 2 SOLUTION

♦ Q1: HTML List

```
<ul>
  <li>Tea
    <ul>
      <li>Hot Tea</li>
      <li>Black Tea</li>
    </ul>
  </li>
  <li>Coffee</li>
  <li>Cold Coffee</li>
  <li>Hot Coffee</li>
</ul>
```

♦ Q2: Employee Form + Display

Form

```
<form method="post" action="emp.php">
Name: <input type="text" name="name"><br>
Salary: <input type="text" name="salary"><br>
Designation: <input type="text" name="desg"><br>
Address: <textarea name="addr"></textarea><br>
<input type="submit">
</form>
```

PHP (emp.php)

```
<?php
echo "Name: ".$_POST['name']. "<br>";
echo "Salary: ".$_POST['salary']. "<br>";
echo "Designation: ".$_POST['desg']. "<br>";
echo "Address: ".$_POST['addr'];
?>
```

SLIP 3 SOLUTION

♦ Q1: PHP Inheritance (Shape)

```
<?php
class Shape {
    function area() {}
}

class Triangle extends Shape {
    function area($b,$h){
        return 0.5*$b*$h;
    }
}

class Square extends Shape {
    function area($s){
        return $s*$s;
    }
}

$t = new Triangle();
echo "Triangle Area: ".$t->area(5,4)."<br>";

$s = new Square();
echo "Square Area: ".$s->area(5);
?>
```

♦ Q2: String Operations

```
<?php
$name = $_POST['name'];

echo "Uppercase: ".strtoupper($name)."<br>";
echo "First Letter Capital: ".ucfirst($name);
?>
```

SLIP 4 SOLUTION

Q.1) PHP Script to Generate XML

```
<?php
$xml = new DOMDocument("1.0", "ISO-8859-1");

// Root element <BookStore>
$bookstore = $xml->createElement("BookStore");
$xml->appendChild($bookstore);

// <Books>
$books = $xml->createElement("Books");
$bookstore->appendChild($books);

// First Book
$php1 = $xml->createElement("PHP");

$title1 = $xml->createElement("Title", "Programming in PHP");
$pub1 = $xml->createElement("Publication", "O'RELLY");

$php1->appendChild($title1);
$php1->appendChild($pub1);

// Second Book
$php2 = $xml->createElement("PHP");

$title2 = $xml->createElement("Title", "Beginners PHP");
$pub2 = $xml->createElement("Publication", "WORX");

$php2->appendChild($title2);
$php2->appendChild($pub2);

// Append to Books
$books->appendChild($php1);
$books->appendChild($php2);

// Display XML
$xml->formatOutput = true;
echo "<pre>";
echo $xml->saveXML();
?>
```

☒ Q.2) PHP Script Using Session (3 Pages)

◇ Page 1: empl.php (Employee Details Form)

```
<form method="post" action="emp2.php">
Employee No: <input type="text" name="eno"><br>
```

```
Employee Name: <input type="text" name="ename"><br>
Address: <textarea name="addr"></textarea><br>
<input type="submit" value="Next">
</form>
```

◇ Page 2: emp2.php (Earnings Form)

```
<?php
session_start();

$_SESSION['eno'] = $_POST['eno'];
$_SESSION['ename'] = $_POST['ename'];
$_SESSION['addr'] = $_POST['addr'];
?>

<form method="post" action="emp3.php">
Basic: <input type="text" name="basic"><br>
DA: <input type="text" name="da"><br>
HRA: <input type="text" name="hra"><br>
<input type="submit" value="Calculate">
</form>
```

◇ Page 3: emp3.php (Display All Details)

```
<?php
session_start();

$eno = $_SESSION['eno'];
$ename = $_SESSION['ename'];
$addr = $_SESSION['addr'];

$basic = $_POST['basic'];
$da = $_POST['da'];
$hra = $_POST['hra'];

$total = $basic + $da + $hra;

echo "<h3>Employee Details</h3>";
echo "Employee No: $eno <br>";
echo "Name: $ename <br>";
echo "Address: $addr <br><br>";

echo "<h3>Salary Details</h3>";
echo "Basic: $basic <br>";
echo "DA: $da <br>";
echo "HRA: $hra <br>";
echo "Total Salary: $total";
?>
```

SLIP 5 SOLUTION

Q.1) Admission Form Design (HTML)

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Admission Form</title>
```

```
</head>
```

```
<body>
```

```
<h2>Admission Form</h2>
```

```
<form>
```

Name: <input type="text" name="name">

Address: <textarea name="address"></textarea>

Gender:

<input type="radio" name="gender" value="Male"> Male

<input type="radio" name="gender" value="Female"> Female

Hobbies:

<input type="checkbox" name="hobby[]" value="Sports"> Sports

<input type="checkbox" name="hobby[]" value="Music"> Music

<input type="checkbox" name="hobby[]" value="Reading"> Reading


```
<h3>Academic Record</h3>
```

```
<table border="1">
```

```
<tr>
```

```
<th>Exam</th>
```

```
<th>Board</th>
```

```
<th>Percentage</th>
```

```
</tr>
```

```
<tr>
```

```
<td>10th</td>
```

```
<td>State Board</td>
```

```
<td>85%</td>
```

```
</tr>
```

```
<tr>
```

```
<td>12th</td>
```

```
<td>State Board</td>
```

```
<td>78%</td>
```

```
</tr>
```

```
<tr>
```

```
<td>Graduation</td>
```

```
<td>University</td>
```

```
<td>82%</td>
```

```
</tr>
```

```
</table><br>

<input type="submit" value="Submit">

</form>

</body>
</html>
```

Q.2) PHP Arithmetic Operation with Default Parameter

```
◊ Page 1: form.html
<!DOCTYPE html>
<html>
<body>

<h2>Arithmetic Operation</h2>

<form method="post" action="result.php">
Number 1: <input type="text" name="num1"><br><br>
Number 2: <input type="text" name="num2"><br><br>

Operation:
<input type="radio" name="op" value="+"> Addition
<input type="radio" name="op" value="-"> Subtraction
<input type="radio" name="op" value="*"> Multiplication
<input type="radio" name="op" value="/"> Division <br><br>

<input type="submit" value="Calculate">

</form>

</body>
</html>
```

```
◊ Page 2: result.php
<?php

// Function with default parameters
function calculate($a = 0, $b = 0, $op = '+')
{
    switch($op)
    {
        case '+': return $a + $b;
        case '-': return $a - $b;
        case '*': return $a * $b;
        case '/': return ($b != 0) ? $a / $b : "Division by zero not allowed";
        default: return "Invalid Operation";
    }
}
}
```

```
// Fetch values
$a = $_POST['num1'];
$b = $_POST['num2'];
$op = $_POST['op'];

// Call function
$result = calculate($a, $b, $op);

// Display result
echo "<h3>Result = $result</h3>";

?>
```

SLIP 6 SOLUTION

Q.1) PHP Script to Display Source Code of a Webpage

PHP provides a built-in function `show_source()` or `highlight_file()`.

```
<?php
// Display source code of current file
show_source("source.php");

// OR you can use:
# highlight_file("source.php");
?>

    ◇ If you want user input:
<form method="post">
Enter file name: <input type="text" name="file">
<input type="submit">
</form>

<?php
if(isset($_POST['file'])){
    show_source($_POST['file']);
}
?>
```

Q.2) PHP Script to Calculate Factorial

◇ Step 1: HTML Form

```
<form method="post" action="fact.php">
Enter Number: <input type="text" name="num">
<input type="submit" value="Find Factorial">
</form>
```

◇ Step 2: PHP Script (fact.php)

```
<?php
```

```
// Function to calculate factorial
function factorial($n)
{
    if($n == 0 || $n == 1)
        return 1;
    else
        return $n * factorial($n - 1);
}
```

```
// Get input
$num = $_POST['num'];
```

```
// Check non-negative
if($num < 0)
```

```
{
  echo "Factorial not defined for negative numbers";
}
else
{
  $result = factorial($num);
  echo "Factorial of $num = $result";
}

?>
```

SLIP 7 SOLUTION

Q.1) PHP Calculator Class

```
<?php
```

```
class Calculator
{
    public $a, $b;

    // Constructor to initialize values
    function __construct($x, $y)
    {
        $this->a = $x;
        $this->b = $y;
    }

    function add()
    {
        return $this->a + $this->b;
    }

    function subtract()
    {
        return $this->a - $this->b;
    }

    function multiply()
    {
        return $this->a * $this->b;
    }

    function divide()
    {
        if($this->b == 0)
            return "Division by zero not allowed";
        return $this->a / $this->b;
    }
}

// Object creation
$c = new Calculator(10, 5);

echo "Addition: " . $c->add() . "<br>";
echo "Subtraction: " . $c->subtract() . "<br>";
echo "Multiplication: " . $c->multiply() . "<br>";
echo "Division: " . $c->divide();
```

?>

Q.2) External CSS Program

- ◇ Step 1: Create CSS file (style.css)

```
body {  
    background-color: skyblue;  
    color: blue;  
    font-size: 18px;  
}
```

-
- ◇ Step 2: Link CSS to HTML file

```
<!DOCTYPE html>  
<html>  
<head>  
    <title>External CSS Example</title>  
    <link rel="stylesheet" type="text/css" href="style.css">  
</head>  
<body>  
  
<h2>This is Web Technology Page</h2>  
<p>This text is displayed in blue color with sky blue background.</p>  
  
</body>  
</html>
```

SLIP 8 SOLUTION

Q.1) PHP Class: Temperature Conversion

◊ Page 1: form.html

```
<!DOCTYPE html>
<html>
<body>

<h2>Temperature Conversion</h2>

<form method="post" action="convert.php">
Enter Celsius: <input type="text" name="c"><br><br>
Enter Fahrenheit: <input type="text" name="f"><br><br>
<input type="submit" value="Convert">
</form>

</body>
</html>
```

◊ Page 2: convert.php

```
<?php

class Temperature
{
    public $celsius;
    public $fahrenheit;

    // Parameterized Constructor
    function __construct($c, $f)
    {
        $this->celsius = $c;
        $this->fahrenheit = $f;
    }

    // Celsius to Fahrenheit
    function cToF()
    {
        return ($this->celsius * 9/5) + 32;
    }

    // Fahrenheit to Celsius
    function fToC()
    {
        return ($this->fahrenheit - 32) * 5/9;
    }
}
```

```
// Get values
$c = $_POST['c'];
$f = $_POST['f'];

// Create object
$temp = new Temperature($c, $f);

// Display result
echo "<h3>Conversion Result</h3>";
echo "Celsius to Fahrenheit: " . $temp->cToF() . "<br>";
echo "Fahrenheit to Celsius: " . $temp->fToC();

?>
```

Q.2) XML + CSS Formatting

```
◇ Step 1: Create XML file (University.xml)
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/css" href="style.css"?>

<Universities>

<Univ>
  <Uname>Savitribai Phule Pune University</Uname>
  <CITY>Pune</CITY>
  <Rank>1</Rank>
</Univ>

<Univ>
  <Uname>Mumbai University</Uname>
  <CITY>Mumbai</CITY>
  <Rank>2</Rank>
</Univ>

<Univ>
  <Uname>Delhi University</Uname>
  <CITY>Delhi</CITY>
  <Rank>3</Rank>
</Univ>

</Universities>
```

```
◇ Step 2: Create CSS file (style.css)
Uname {
  color: black;
  font-family: "Copperplate Gothic Light";
  font-size: 16pt;
  font-weight: bold;
  display: block;
```

```
}
```

```
CITY, Rank {  
  color: yellow;  
  font-family: Arial;  
  font-size: 12pt;  
  font-weight: bold;  
  display: block;  
}
```

```
Univ {  
  margin-bottom: 15px;  
  display: block;  
}
```

SLIP 9 SOLUTION

Q.1) JavaScript to Print Table of First n Numbers

```
<!DOCTYPE html>
<html>
<body>

<h2>Tables of First n Numbers</h2>

Enter n: <input type="number" id="num">
<button onclick="printTable()">Print Tables</button>

<pre id="output"></pre>

<script>
function printTable()
{
    var n = document.getElementById("num").value;
    var result = "";

    for(let i = 1; i <= n; i++)
    {
        result += "Table of " + i + "\n";
        for(let j = 1; j <= 10; j++)
        {
            result += i + " x " + j + " = " + (i*j) + "\n";
        }
        result += "\n";
    }

    document.getElementById("output").innerHTML = result;
}
</script>

</body>
</html>
```

Q.2) PHP Menu Driven Program (Associative Arrays)

```
<?php

// Sample associative arrays
$arr1 = array("a"=>10, "b"=>20, "c"=>30);
$arr2 = array("b"=>20, "c"=>40, "d"=>50);

// Menu
echo "<h3>Menu</h3>";
echo "1. Merge Arrays<br>";
echo "2. Intersection<br>";
echo "3. Union<br>";
```

```

echo "4. Difference<br><br>";

// Choice (you can change value for testing)
$choice = 1;

switch($choice)
{
    case 1:
        echo "<b>Merged Array:</b><br>";
        print_r(array_merge($arr1, $arr2));
        break;

    case 2:
        echo "<b>Intersection:</b><br>";
        print_r(array_intersect_assoc($arr1, $arr2));
        break;

    case 3:
        echo "<b>Union:</b><br>";
        $union = $arr1 + $arr2;
        print_r($union);
        break;

    case 4:
        echo "<b>Difference (arr1 - arr2):</b><br>";
        print_r(array_diff_assoc($arr1, $arr2));
        break;

    default:
        echo "Invalid Choice";
}

?>
Alternative (User Input Menu Version – Better for Exam)
<form method="post">
Enter Choice (1-4): <input type="text" name="ch">
<input type="submit">
</form>

<?php
$arr1 = array("a"=>10, "b"=>20, "c"=>30);
$arr2 = array("b"=>20, "c"=>40, "d"=>50);

if(isset($_POST['ch']))
{
    $ch = $_POST['ch'];

    switch($ch)
    {
        case 1:

```

```
print_r(array_merge($arr1, $arr2));  
break;
```

case 2:

```
print_r(array_intersect_assoc($arr1, $arr2));  
break;
```

case 3:

```
print_r($arr1 + $arr2);  
break;
```

case 4:

```
print_r(array_diff_assoc($arr1, $arr2));  
break;
```

```
}
```

```
}  
?>
```

SLIP 10 SOLUTION

Q.1) PHP Inheritance (Shape with Menu using Radio Button)

◇ HTML Form

```
<form method="post">
Select Shape:<br>
<input type="radio" name="shape" value="triangle"> Triangle<br>
<input type="radio" name="shape" value="square"> Square<br>
<input type="radio" name="shape" value="rectangle"> Rectangle<br>
<input type="radio" name="shape" value="circle"> Circle<br><br>

Enter Value 1: <input type="text" name="v1"><br><br>
Enter Value 2 (if needed): <input type="text" name="v2"><br><br>

<input type="submit" value="Calculate Area">
</form>
```

◇ PHP Script

```
<?php

// Base Class
class Shape {
    function area() {
        return 0;
    }
}

// Triangle
class Triangle extends Shape {
    function area($b, $h) {
        return 0.5 * $b * $h;
    }
}

// Square
class Square extends Shape {
    function area($s) {
        return $s * $s;
    }
}

// Rectangle
class Rectangle extends Shape {
    function area($l, $w) {
        return $l * $w;
    }
}
```

```
// Circle
class Circle extends Shape {
    function area($r) {
        return 3.14 * $r * $r;
    } // Processing
}
if(isset($_POST['shape'])) {
    $shape = $_POST['shape'];
    $v1 = $_POST['v1'];
    $v2 = $_POST['v2'];

    switch($shape) {
        case "triangle":
            $obj = new Triangle();
            echo "Area of Triangle = " . $obj->area($v1, $v2);
            break;

        case "square":
            $obj = new Square();
            echo "Area of Square = " . $obj->area($v1);
            break;

        case "rectangle":
            $obj = new Rectangle();
            echo "Area of Rectangle = " . $obj->area($v1, $v2);
            break;

        case "circle":
            $obj = new Circle();
            echo "Area of Circle = " . $obj->area($v1);
            break;
    }
}
?>
```

Q.2) PHP Superglobal Variables

```
<?php
```

```
// a) Client IP Address
```

```
echo "<b>Client IP Address:</b> " . $_SERVER['REMOTE_ADDR'] . "<br><br>";
```

```
// b) Browser Information
```

```
echo "<b>Browser Information:</b> " . $_SERVER['HTTP_USER_AGENT'] . "<br><br>";
```

```
// c) Check HTTP or HTTPS
```

```
if(isset($_SERVER['HTTPS']) && $_SERVER['HTTPS'] == 'on')
```

```
    echo "<b>Page loaded using HTTPS</b>";
```

```
else
```

```
    echo "<b>Page loaded using HTTP</b>";
```

```
?>
```

SLIP 11 SOLUTION

Q.1) JavaScript to Calculate Simple Interest

```
<!DOCTYPE html>
<html>
<body>

<h2>Simple Interest Calculator</h2>

Principal: <input type="text" id="p"><br><br>
Rate (%): <input type="text" id="r"><br><br>
Time (years): <input type="text" id="t"><br><br>

<button onclick="calculateSI()">Calculate</button>

<p id="result"></p>

<script>
function calculateSI()
{
    var p = parseFloat(document.getElementById("p").value);
    var r = parseFloat(document.getElementById("r").value);
    var t = parseFloat(document.getElementById("t").value);

    var si = (p * r * t) / 100;

    document.getElementById("result").innerHTML = "Simple Interest = " + si;
}
</script>

</body>
</html>
```

Q.2) AJAX Program to Read contact.dat and Display in Table

```
◇ Step 1: Create contact.dat file
1,John,0201234567,9876543210,Friend
2,Rahul,0207654321,9123456780,Colleague
3,Anita,0201112222,9988776655,Sister

◇ Step 2: HTML + AJAX
<!DOCTYPE html>
<html>
<body>

<h2>Contact List</h2>

<button onclick="loadData()">Print Contacts</button>
```

```
<div id="table"></div>
```

```
<script>
```

```
function loadData()
```

```
{
```

```
    var xhttp = new XMLHttpRequest();
```

```
    xhttp.onreadystatechange = function()
```

```
    {
```

```
        if (this.readyState == 4 && this.status == 200)
```

```
        {
```

```
            var data = this.responseText.trim().split("\n");
```

```
            var table = "<table border='1'>";
```

```
            table +=
```

```
"<tr><th>SrNo</th><th>Name</th><th>Residence</th><th>Mobile</th><th>Relation</th></tr>";
```

```
            for (var i = 0; i < data.length; i++)
```

```
            {
```

```
                var row = data[i].split(",");
```

```
                table += "<tr>";
```

```
                for (var j = 0; j < row.length; j++)
```

```
                {
```

```
                    table += "<td>" + row[j] + "</td>";
```

```
                }
```

```
                table += "</tr>";
```

```
            }
```

```
            table += "</table>";
```

```
            document.getElementById("table").innerHTML = table;
```

```
        }
```

```
    };
```

```
    xhttp.open("GET", "contact.dat", true);
```

```
    xhttp.send();
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

SLIP 12 SOLUTION

Q.1) Divide Screen into Four Equal Parts

Using HTML + CSS (frames/grid layout)

```
<!DOCTYPE html>
<html>
<head>
<style>
.container {
  display: grid;
  grid-template-columns: 50% 50%;
  grid-template-rows: 50% 50%;
  height: 100vh;
}

.box {
  border: 1px solid black;
  padding: 10px;
}
</style>
</head>

<body>

<div class="container">

<div class="box">
<h3>Technical Activities</h3>
<ul>
<li>Workshops</li>
<li>Seminars</li>
<li>Guest Lectures</li>
</ul>
</div>

<div class="box">
<h3>Cultural Activities</h3>
<ul>
<li>Dance</li>
<li>Drama</li>
<li>Music</li>
</ul>
</div>

<div class="box">
<h3>Sports Activities</h3>
<ul>
<li>Cricket</li>
```

```

<li>Football</li>
<li>Badminton</li>
</ul>
</div>

<div class="box">
<h3>Academic Activities</h3>
<ul>
<li>Projects</li>
<li>Assignments</li>
<li>Presentations</li>
</ul>
</div>

</div>

</body>
</html>

```

Q.2) PHP Cookie Program (3 Pages)

◊ Page 1: form.html (User Preferences)

```

<form method="post" action="setcookie.php">

Font Style: <input type="text" name="style"><br><br>
Font Size: <input type="text" name="size"><br><br>
Font Color: <input type="text" name="color"><br><br>
Background Color: <input type="text" name="bg"><br><br>

<input type="submit" value="Save Preferences">

</form>

```

◊ Page 2: setcookie.php (Store in Cookies)

```

<?php

setcookie("style", $_POST['style'], time()+3600);
setcookie("size", $_POST['size'], time()+3600);
setcookie("color", $_POST['color'], time()+3600);
setcookie("bg", $_POST['bg'], time()+3600);

echo "Preferences Saved!<br>";
echo "<a href='display.php'>View Preferences</a>";

?>

```

◊ Page 3: display.php (Apply Preferences)

```

<?php
$style = $_COOKIE['style'];
$size = $_COOKIE['size'];

```

```
$color = $_COOKIE['color'];  
$bg = $_COOKIE['bg'];  
?>
```

```
<html>  
<body style="  
    font-family: <?php echo $style; ?>;  
    font-size: <?php echo $size; ?>px;  
    color: <?php echo $color; ?>;  
    background-color: <?php echo $bg; ?>;  
">
```

```
<h2>Applied User Preferences</h2>
```

```
<?php  
echo "Font Style: $style <br>";  
echo "Font Size: $size <br>";  
echo "Font Color: $color <br>";  
echo "Background Color: $bg <br>";  
?>
```

```
</body>  
</html>
```

SLIP 13 SOLUTION

Q.1) Java Program – Check Voting Eligibility

```
import java.util.Scanner;

public class VotingEligibility {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Age: ");
        int age = sc.nextInt();

        if(age >= 18) {
            System.out.println("Candidate is Eligible for Voting");
        } else {
            System.out.println("Candidate is NOT Eligible for Voting");
        }

        sc.close();
    }
}
```

Q.2) Java Program – Store n Integers in ArrayList

```
import java.util.ArrayList;
import java.util.Scanner;

public class ArrayListExample {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        ArrayList<Integer> list = new ArrayList<>();

        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();

        // Accept n integers
        for(int i = 0; i < n; i++) {
            System.out.print("Enter element " + (i+1) + ": ");
            int num = sc.nextInt();
            list.add(num);
        }

        // Display elements
        System.out.println("\nElements in ArrayList:");
        for(int i = 0; i < list.size(); i++) {
            System.out.println(list.get(i));
        }
        sc.close();
    }
}
```

SLIP 14 SOLUTION

Q.1) JavaScript – Display Weekday from Number

```
<!DOCTYPE html>
<html>
<body>

<h2>Display Weekday</h2>

Enter number (1-7): <input type="number" id="day">
<button onclick="showDay()">Show Day</button>

<p id="result"></p>

<script>
function showDay()
{
    var d = parseInt(document.getElementById("day").value);
    var dayName;

    switch(d)
    {
        case 1: dayName = "Monday"; break;
        case 2: dayName = "Tuesday"; break;
        case 3: dayName = "Wednesday"; break;
        case 4: dayName = "Thursday"; break;
        case 5: dayName = "Friday"; break;
        case 6: dayName = "Saturday"; break;
        case 7: dayName = "Sunday"; break;
        default: dayName = "Invalid Input";
    }

    document.getElementById("result").innerHTML = dayName;
}
</script>

</body>
</html>
```

Q.2) Database + PHP Solution

◇ a) Relationship (Many-to-Many)

A student can participate in many competitions

A competition can have many students

So we create a junction table:

Student (stud_id, name, class)

Competition (c_no, c_name, type)

Participation (stud_id, c_no, rank, year)

◇ b) RDB in 3NF (SQL)

```
CREATE TABLE Student (  
    stud_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    class VARCHAR(20)  
);
```

```
CREATE TABLE Competition (  
    c_no INT PRIMARY KEY,  
    c_name VARCHAR(50),  
    type VARCHAR(20)  
);
```

```
CREATE TABLE Participation (  
    stud_id INT,  
    c_no INT,  
    rank INT,  
    year INT,  
    PRIMARY KEY (stud_id, c_no),  
    FOREIGN KEY (stud_id) REFERENCES Student(stud_id),  
    FOREIGN KEY (c_no) REFERENCES Competition(c_no)  
);
```

◇ c) PHP Script (Display 1st Rank Students)

HTML Form

```
<form method="post" action="result.php">  
Enter Competition Name: <input type="text" name="cname">  
<input type="submit" value="Search">  
</form>
```

PHP (result.php)

```
<?php
```

```
$conn = new mysqli("localhost", "root", "", "test");
```

```
$cname = $_POST['cname'];
```

```
$sql = "SELECT s.stud_id, s.name, s.class  
        FROM Student s, Competition c, Participation p  
        WHERE s.stud_id = p.stud_id  
        AND c.c_no = p.c_no  
        AND p.rank = 1  
        AND c.c_name = '$cname'";
```

```
$result = $conn->query($sql);
```

```
echo "<h3>Students with 1st Rank</h3>";
```

```
if($result->num_rows > 0)  
{
```

```
while($row = $result->fetch_assoc())
{
    echo "ID: ".$row['stud_id']." Name: ".$row['name']." Class: ".$row['class']."<br>";
}
else
{
    echo "No records found";
}

$conn->close();

?>
```

SLIP 15 SOLUTION

Q.1) JavaScript – Check Prime Number

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>Prime Number Check</h2>
```

```
Enter Number: <input type="number" id="num">
```

```
<button onclick="checkPrime()">Check</button>
```

```
<p id="result"></p>
```

```
<script>
```

```
function checkPrime()
```

```
{
```

```
    var n = parseInt(document.getElementById("num").value);
```

```
    var flag = 1;
```

```
    if(n <= 1)
```

```
        flag = 0;
```

```
    else {
```

```
        for(var i = 2; i <= n/2; i++)
```

```
        {
```

```
            if(n % i == 0)
```

```
            {
```

```
                flag = 0;
```

```
                break;
```

```
            }
```

```
        }
```

```
    }
```

```
    if(flag == 1)
```

```
        document.getElementById("result").innerHTML = n + " is Prime Number";
```

```
    else
```

```
        document.getElementById("result").innerHTML = n + " is NOT Prime Number";
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

Q.2) AJAX – Search Student Name (Using Array)

◇ HTML + JavaScript (AJAX Simulation)

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>Search Student</h2>
```

```
Enter Name: <input type="text" onkeyup="searchName(this.value)">
```

```
<ul id="result"></ul>
```

```
<script>
```

```
// Student array
```

```
var students = ["Amit", "Rahul", "Ramesh", "Sneha", "Sonal", "Anita", "Akash"];
```

```
function searchName(str)
```

```
{  
    var res = "";  
  
    if(str.length == 0)  
    {  
        document.getElementById("result").innerHTML = "";  
        return;  
    }  
}
```

```
// Simulating AJAX filtering
```

```
for(var i = 0; i < students.length; i++)
```

```
{  
    if(students[i].toLowerCase().startsWith(str.toLowerCase()))  
    {  
        res += "<li>" + students[i] + "</li>";  
    }  
}
```

```
document.getElementById("result").innerHTML = res;  
}
```

```
</script>
```

```
</body>
```

```
</html>
```

Alternative (Real AJAX using separate file)

◇ HTML

```
<input type="text" onkeyup="showHint(this.value)">
```

```
<p id="txtHint"></p>
```

```
<script>
```

```
function showHint(str)
```

```
{  
    var xmlhttp = new XMLHttpRequest();  
  
    xmlhttp.onreadystatechange = function()
```

```

    {
        if(this.readyState == 4 && this.status == 200)
        {
            document.getElementById("txtHint").innerHTML = this.responseText;
        }
    };

    xhttp.open("GET", "search.php?q=" + str, true);
    xhttp.send();
}
</script>

```

◇ PHP (search.php)

```
<?php
```

```
$students = array("Amit","Rahul","Ramesh","Sneha","Sonal","Anita","Akash");
```

```
$q = $_GET["q"];
```

```
$result = "";
```

```
foreach($students as $name)
```

```

{
    if(strpos($name, $q) === 0)
    {
        $result .= $name . "<br>";
    }
}

```

```
echo ($result == "") ? "No match found" : $result;
```

```
?>
```